

Exit Ticket Questions for Week 2

1. In one or two sentences, explain why sampling creates spectral replicas.
2. Explain in one sentence why ideal reconstruction produces "sinc" interpolation.

Good answers:

1. Sampling multiplies the signal by a periodic impulse train. Multiplication in time becomes convolution in frequency, creating shifted copies of the original spectrum at multiples of the sampling frequency.

Look for the mathematical mechanism, periodic impulse train, multiplication in one domain is convolution in the other domain.

2. Because the ideal reconstruction system is a brick-wall lowpass filter whose impulse response is a "sinc".

Look for ideal reconstruction by rectangular lowpass filter in frequency domain and rectangular $\Leftrightarrow$ sinc relation between frequency and time domains.

A good exit ticket answer should be short, correct, mathematically sound, to the point and demonstrate a well-reasoned understanding of the subject that the question is asking about. Exit ticket answers will be graded on a scale of “low” (1 point), “medium” (2 points), or “high” (3 points).

Use LaTeX style for math expressions, e.g., $\delta[n]$ for a discrete-time impulse.